

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

— * —



BÁO CÁO BÀI TẬP

Chương 2: Lập Trình Socket Cơ Bản

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Link github : <https://github.com/DinhMinhChien/Lap-trinh-mang>

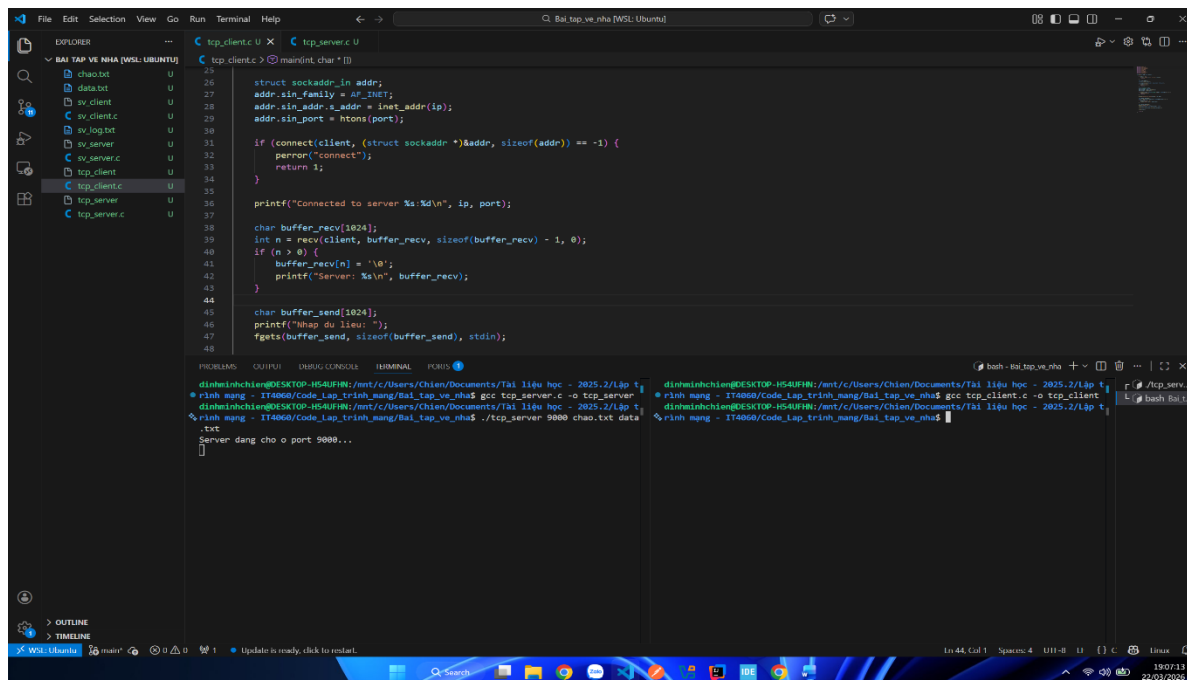
Bài 1 – 2 : TCP_client & TCP_server

a) TCP_server

- Command :

```
gcc tcp_server.c -o tcp_server
./tcp_server 9000 chao.txt data.txt
```

- Địa chỉ 127.0.0.1:9000



- File để lấy message gửi cho client : chao.txt

The screenshot shows the VS Code interface with the Explorer view on the left displaying a file tree for 'BAI_TAP_VE_NHA'. The file 'chao.txt' is selected. The main editor shows the content of 'chao.txt' with two lines: '1 Xin chao client!' and '2'. The bottom panel shows the TERMINAL view with two terminal windows. The left terminal window shows the command 'gcc tcp_server.c -o tcp_server' and the output 'Server dang cho o port 9000...'. The right terminal window shows the command './tcp_server 9000 chao.txt data.txt' and the output 'Server dang cho o port 9000...'.

- File để lưu message do client gửi đến : data.txt

The screenshot shows the VS Code interface with the Explorer view on the left displaying a file tree for 'BAI_TAP_VE_NHA'. The file 'data.txt' is selected. The main editor shows the content of 'data.txt' with two lines: '1' and '2'. The bottom panel shows the TERMINAL view with two terminal windows. The left terminal window shows the command 'gcc tcp_client.c -o tcp_client' and the output 'Server dang cho o port 9000...'. The right terminal window shows the command './tcp_client 127.0.0.1 9000' and the output 'Server dang cho o port 9000...'.

b) TCP_client :

- Command :

```
gcc tcp_client.c -o tcp_client
./tcp_client 127.0.0.1 9000
```
- Client lắng nghe trên kết nối đến server 127.0.0.1:9000

- Kết nối thành công in ra “Connected to server 127.0.0.1:9000”
- Server gửi message là “Xin chào client!” như trong file chào.txt và client in ra

```

File Edit Selection View Go Run Terminal Help
BÀI TẬP VỀ NHÀ (WSL: Ubuntu)
data.txt
1
2
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_server.c -o tcp_server
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_server 9000 chào.txt data
.txt
Server đang chờ ở port 9000...
Có client kết nối!

dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_client.c -o tcp_client
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_client 127.0.0.1 9000
Connected to server 127.0.0.1:9000
Server: Xin chào client!

Nhập dữ liệu:

```

- Client gửi “client1 – server” cho server và được ghi vào file data.txt

```

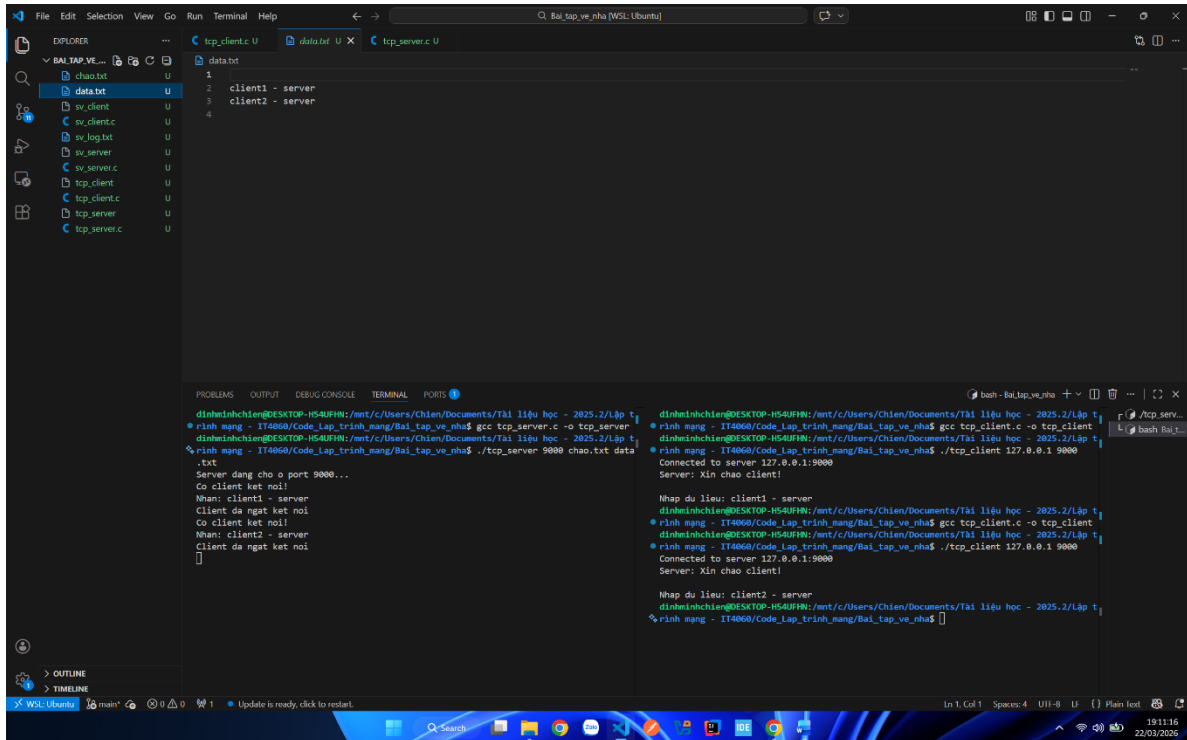
File Edit Selection View Go Run Terminal Help
BÀI TẬP VỀ NHÀ (WSL: Ubuntu)
data.txt
1
2
3
client1 - server
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_server.c -o tcp_server
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_server 9000 chào.txt data
.txt
Server đang chờ ở port 9000...
Có client kết nối!
Nhập: client1 - server
Client đã ngắt kết nối

dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_client.c -o tcp_client
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_client 127.0.0.1 9000
Connected to server 127.0.0.1:9000
Server: Xin chào client!

Nhập dữ liệu: client1 - server
dinhhinhchien@DESKTOP-H54UFIH: /mnt/c/Users/Chien/Documents/Tài liệu học - 2025.2/Lập t
rình mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$

```

- Tương tự với client 2 gửi “client2 – server” cho server và dữ liệu được ghi vào file data.txt



```
File Edit Selection View Go Run Terminal Help
Bai_tap_ve_nha (WSL: Ubuntu)
DOPLOER
BAL_TAP_VE_NHA...
chao.txt
data.txt
sv_client
sv_client.c
sv_log.txt
sv_server
sv_server.c
tcp_client
tcp_client.c
tcp_server
tcp_server.c
data.txt
1
2 client1 - server
3 client2 - server
4

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
• rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_server.c -o tcp_server
dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
% rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_server 9000 chao.txt data
.txt
Server dang cho o port 9000...
Co client ket noi!
Nhan: client1 - server
Client da ngat ket noi!
Co client ket noi!
Nhan: client2 - server
Client da ngat ket noi
[]

dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
• rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_client.c -o tcp_client
dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
• rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_client 127.0.0.1 9000
Connected to server 127.0.0.1:9000
Server: Xin chao client!

Nhap du lieu: client1 - server
dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
• rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ gcc tcp_client.c -o tcp_client
dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
• rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ ./tcp_client 127.0.0.1 9000
Connected to server 127.0.0.1:9000
Server: Xin chao client!

Nhap du lieu: client2 - server
dinhminhchien@DESKTOP-H54UFWN:/mnt/c/Users/Chien/Documents/Tai lieu hoc - 2025.2/Lap t...
% rinh mạng - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha$ []
```

Bài 3-4 : SV_client & SV_server

c) SV_server:

- Command :

```
gcc sv_server.c -o sv_server
./sv_server 9001 sv_log.txt
```

- Địa chỉ : 127.0.0.1:9001

The screenshot shows the VS Code editor with the file explorer on the left displaying a project named 'BAL_TAP_VE_NHA (WSL: UBUNTU)'. The file 'sv_server.c' is open in the editor. The code is a C program for a simple TCP server. It includes `<stdio.h>` and `<time.h>`, and defines a `BUFFER_SIZE` of 1024. The `main` function takes three arguments: `argc` and `argv`. It checks if `argc` is 3, and if so, it prints the usage and returns 1. It then converts the port from `argv[1]` to an integer and sets `logfile` to `argv[2]`. It initializes `server_fd` and `client_fd`, sets up the `server_addr` and `client_addr` structures, and sets the `socklen_t` values. It then creates a socket, binds it to the `server_addr`, and listens on the specified port. A notification at the bottom right states: 'Your application running on port 9001 is available. See all forwarded ports'. The terminal at the bottom shows the command `gcc sv_server.c -o sv_server` and the output `Server listening on port 9001...`.

- File để ghi log : sv_log.txt

The screenshot shows the VS Code editor with the file explorer on the left displaying the same project. The file 'sv_log.txt' is open in the editor. The file is empty. The terminal at the bottom shows the command `gcc sv_server.c -o sv_server` and the output `Server listening on port 9001...`. The status bar at the bottom right shows the file is 'Plain text'.

d) SV_client:

- Command:

```
gcc sv_client.c -o sv_client
./sv_client 127.0.0.1 9001
```

- Kết nối đến server 127.0.0.1:9001

The screenshot shows the Visual Studio Code interface with a C project named 'bai_tap_ve_nha' in a WSL Ubuntu environment. The Explorer pane on the left shows files like 'chao.txt', 'data.txt', 'sv_client.c', 'sv_log.txt', 'sv_server.c', 'tcp_client.c', and 'tcp_server.c'. The main editor displays the code for 'sv_client.c', which includes a `main` function that scans for user input, formats it into a buffer, and sends it to the server using `send`. The bottom terminal pane shows the execution of the server and client programs. The server output indicates it is listening on port 9001. The client output shows it successfully connects to the server at 127.0.0.1:9001 and sends an 'MSV:' message. A notification at the bottom right states 'Your application running on port 9001 is available'.

- Client nhập thông tin vào và gửi tới server

This screenshot shows the same VS Code environment as the previous one, but with the client program running and sending data to the server. The terminal output for the client shows the user inputting their name 'Dinh Minh Chien', date of birth '2005-08-07', and a message 'Da gui du lieu len server!'. The server's output shows it has received this information and formatted it into a log entry: '127.0.0.1 2026-03-22 19:39:21 20235278 Dinh Minh Chien 2005-08-07 3.20'. The client's output also shows the 'MSV:' message being sent.

- Server ghi vào file sv_log.txt đồng thời in trên terminal

The screenshot shows the VS Code interface with a file explorer on the left displaying a project structure for 'BAI_TAP_VE_NHA'. The file 'sv_log.txt' is selected. The main editor area shows the content of 'sv_log.txt', which contains a log entry for a client connection from 127.0.0.1 at 2025-03-22 19:39:21. The terminal window at the bottom shows the execution of a C++ program. The program output includes the client's name 'Dinh Minh Chien', birth date '2005-08-07', and a message 'Da gui du lieu len server!'. The program also shows the client's IP address '127.0.0.1' and the server's IP address '127.0.0.1'.

- Tương tự với một client nữa:

The screenshot shows the VS Code interface with a file explorer on the left displaying a project structure for 'BAI_TAP_VE_NHA'. The file 'sv_log.txt' is selected. The main editor area shows the content of 'sv_log.txt', which contains two log entries. The first entry is for a client connection from 127.0.0.1 at 2025-03-22 19:39:21. The second entry is for a client connection from 127.0.0.1 at 2025-03-22 19:40:36. The terminal window at the bottom shows the execution of a C++ program. The program output includes the client's name 'Dinh Minh Chien', birth date '2005-08-07', and a message 'Da gui du lieu len server!'. The program also shows the client's IP address '127.0.0.1' and the server's IP address '127.0.0.1'.